

Python Programming



Assignment Deadline Closed

- Bank Class



Marks Distribution

**Tentative*

Assignments	15%
Classroom Participation	5%
Project	50%
Exam	30%

Guideline on Project

- Will have a mid-term presentation (after Tihar) where the group has to present what they have done and what things are yet to be implemented.
 - Marks will be given based on github commits, presentation, Q/A *(Further guidelines will be posted)*
 - A final presentation will be on the month of Falgun at an exhibition and final marks will be awarded based on judges and people rating
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Revision Class

- Exam will cover contents till Exception Handling starting from Hello World
 - Till next week prepare a presentation on whatever you have learnt
 - Around 5 lucky students will get to present in the next class.
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Diabetes Prediction Using Machine Learning

a machine learning model capable of predicting whether a patient is likely to have diabetes using health-related data

Leader: Abanish Dhakal



Members: Arman Shrestha, Syed Samad Shah, Aakanchha Kharel

Virtual Drum with hardware visualizer and Sketchboard

Multiverse of IOT

Leader: Anshudeep Amgai



Members: Samir Ghimire

Gesture based musical program

Two-part live instrument

Leader: Niaz Shrestha



Members: Rhythm Adhikari

Fake news / spam detection dataset

Predict and identify if a message/news is fake or real

Leader: Samrajya Bikram Bohara



Members: Saksham Prasai, Samman Sedhai, Saurya Bikram Bisht

A hand gesture based CAD using ml models

3D voxel making using openml

Leader: Gehendraraj Pyakurel



Members: Arpan Chalise, Bivaan Poudel, Aahan Raj Shrestha

Student Grade Management System

Python Project

Leader: Aarjan Shrestha



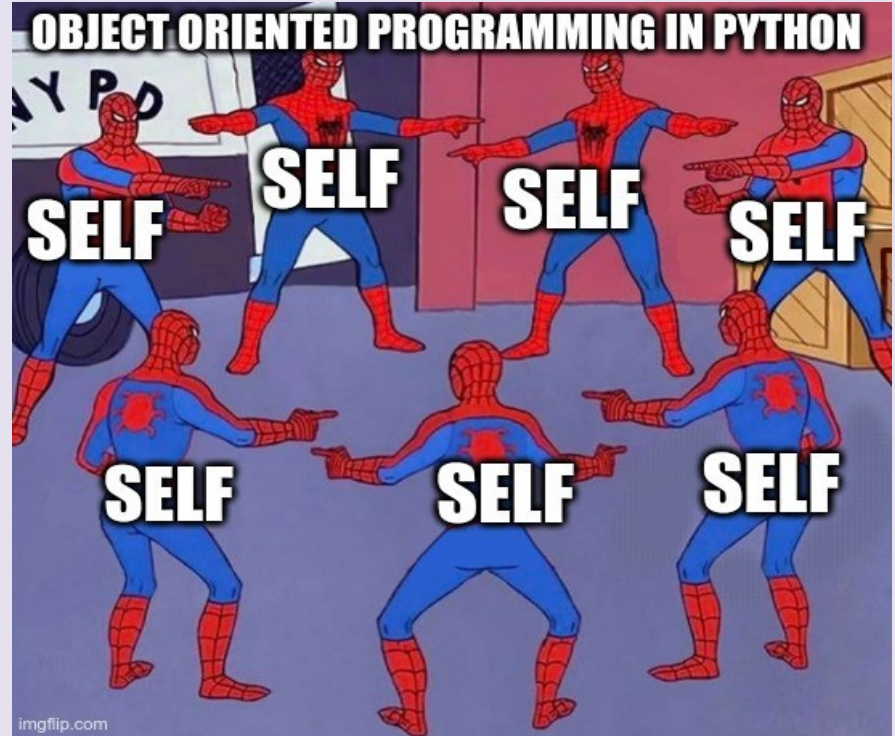
Members: Pranisha Shrestha

Guideline on Exam

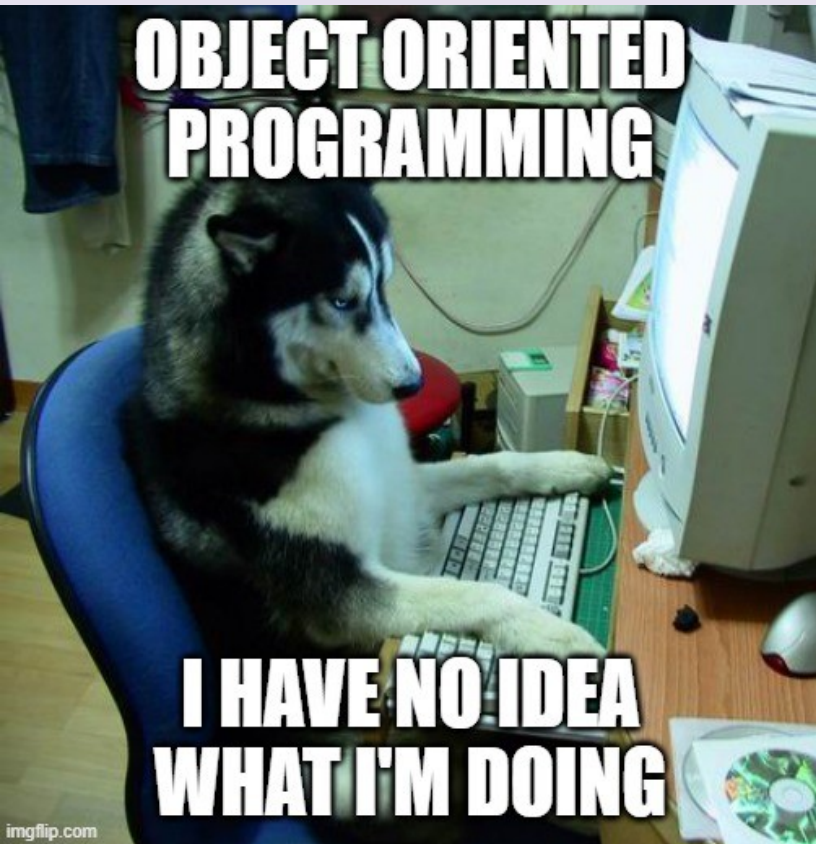
- MCQ
 - Fill in the blanks
 - Write short notes on (mostly from Projects and Assignments)
 - What will the output of the code be?
 - Debug the code.
 - Write program on specified cases
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OOP

- Python is an object-oriented language, allowing you to structure your code using classes and objects for better organization and reusability.
- A software design approach that organizes code around "objects" rather than just actions and logic



Class



The DRY principle means you should avoid writing the same code more than once. Move repeated code into functions or classes and reuse it.

Inheritance

- Inheritance means receiving money, property, traits, or features from someone else.
- This transfer can happen through family death, biological genes, or computer code.
- In Python, OOP allows a new class (child or derived class) to adopt all the methods and properties from an existing class (parent or base class)



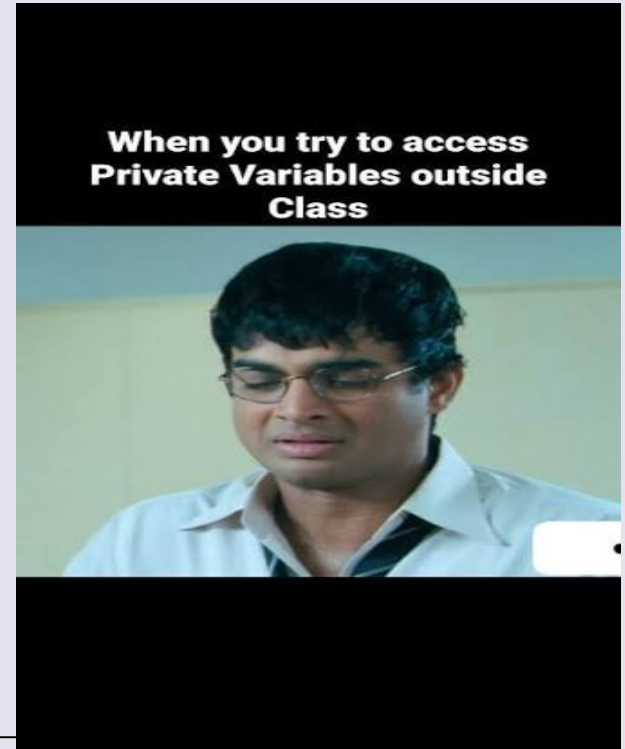
Polymorphism



Polymorphism comes from Greek roots meaning "many forms". It describes how a single word, idea, function, or gene can show up or work in different ways depending on the situation

Encapsulation

- Python, access modifiers (public, protected, and private) are used to implement encapsulation by managing how class variables and methods are accessed.
- Encapsulation is one of the four core pillars of Object-Oriented Programming (OOP).
- It refers to the practice of bundling data (variables) and the methods that operate on that data into a single unit (a class), while restricting direct access to some of the object's components.



Write a OOP Class to mimic bank functionalities



- Deposit Money
- Withdraw Money



- Deposit Money
 - Withdraw Money
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